TASK-3

Exploratory Data Analysis (EDA) on Netflix Dataset

Dataset Source: [Kaggle - Netflix Movies and TV Shows Dataset]

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1.Introduction

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Netflix dataset to identify trends, patterns, and insights from the content available on the platform. The dataset contains information about Netflix movies and TV shows, including title, director, cast, country, release year, rating, duration, genre, and description.

2.Data Cleaning

The dataset was examined for missing values before analysis. The following missing values were identified:

Column	**Missing Values**
Director	2634
Cast	825
Country	831
Date Added	10
Rating	4
Duration	3

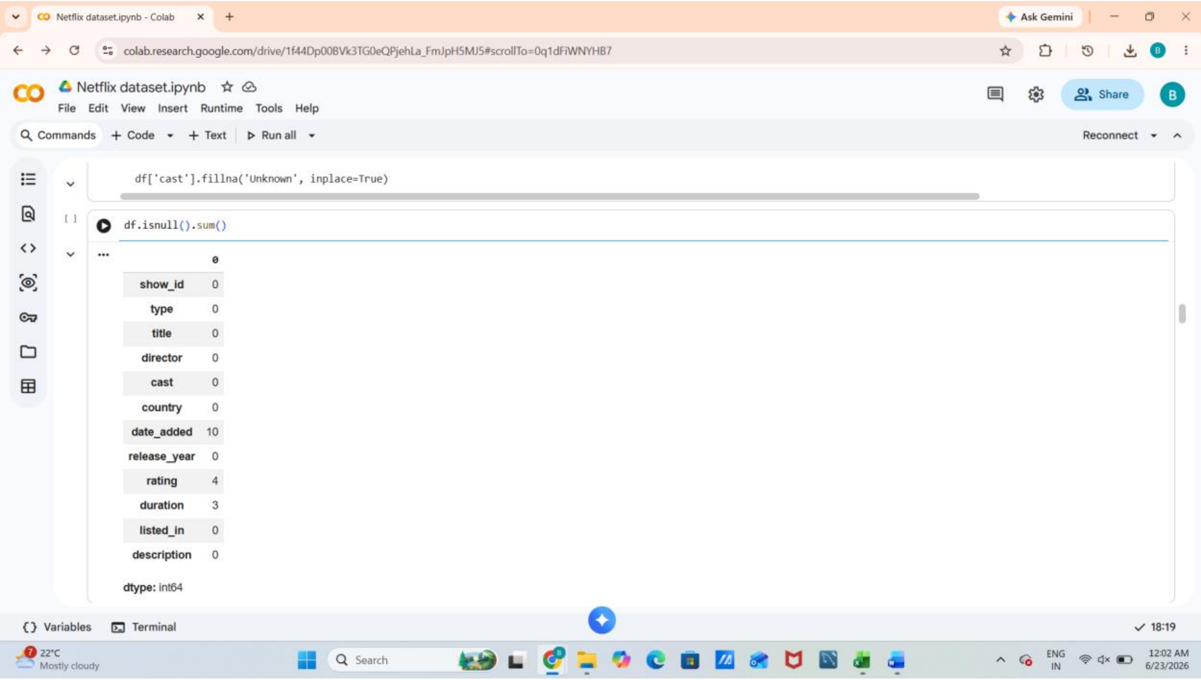
Data Cleaning Steps Performed

1. Missing values in the **Director** column (2634) were replaced with **"Unknown"**.
2. Missing values in the **Cast** column (825) were replaced with **"Unknown"**.
3. Missing values in the **Country** column (831) were replaced with **"Unknown"**.
4. Missing values in the **Date Added** column (10) were filled using the most frequently occurring value (Mode).

5. Missing values in the **Rating** column (4) were filled using the most frequently occurring value (Mode).

6. Missing values in the **Duration** column (3) were filled using the most frequently occurring value (Mode).

After preprocessing, all missing values were successfully handled, resulting in a clean dataset suitable for analysis.



3.Summary Statistics

The Netflix dataset contains 8,807 records and 12 attributes describing movies and TV shows available on Netflix.

Key observations from the dataset include:

1. The dataset contains both Movies and TV Shows.

2. Movies represent a larger proportion of the content library.

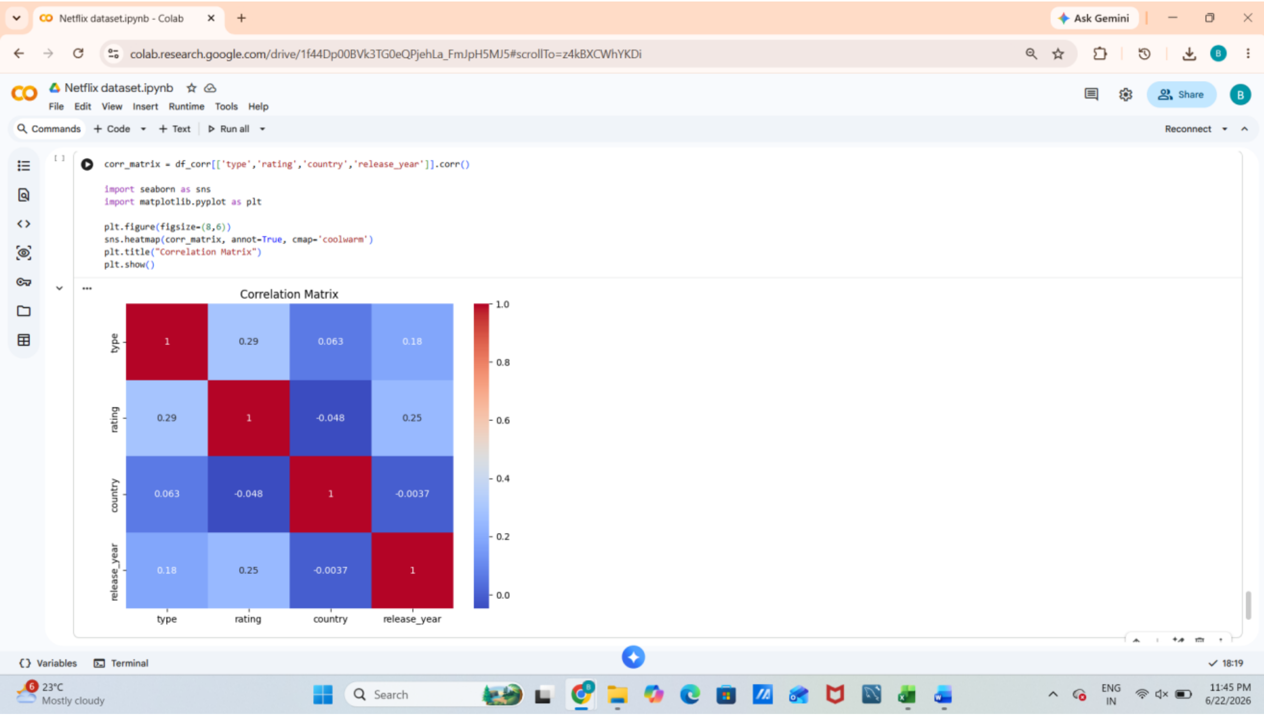
3. Content originates from multiple countries across the world.

4. The dataset covers content released over several decades.

4.Correlation Analysis

To perform correlation analysis, selected categorical variables were converted into numerical values using Label Encoding.

Correlation Heatmap

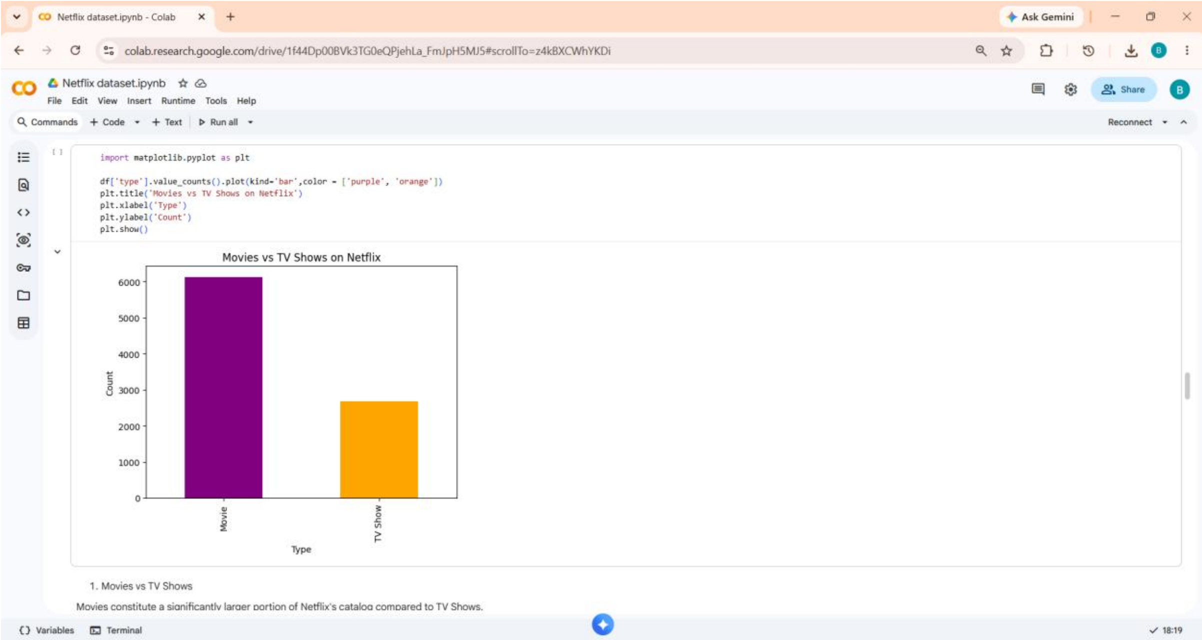


Observation

The dataset primarily consists of categorical variables; therefore, correlation analysis provides limited insights. Encoded variables were used to explore potential relationships between content type, rating, and release year.

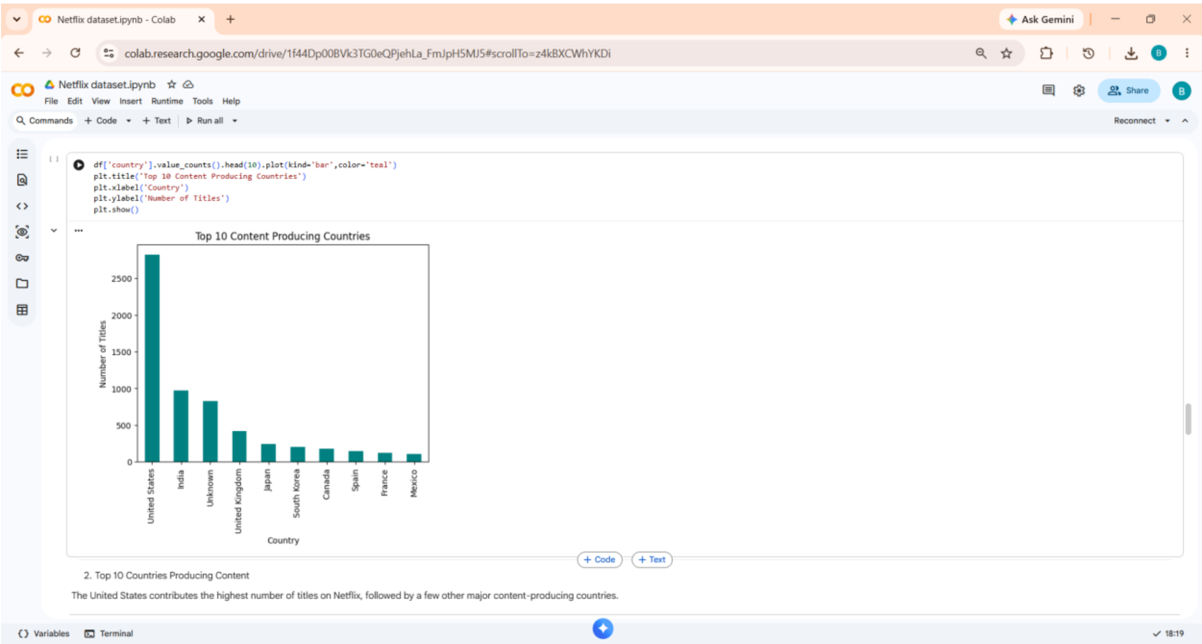
5.Trend Identification and Insights.

1. Movies vs TV Shows



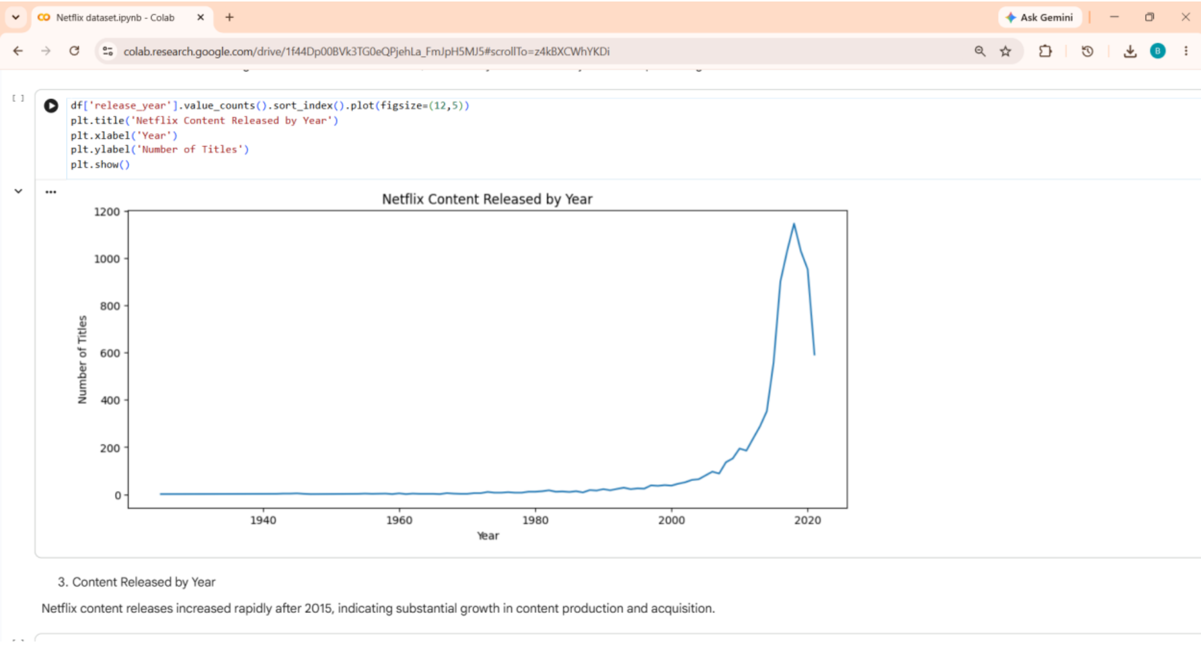
Observation: Movies constitute a significantly larger portion of Netflix's catalog compared to TV Shows.

2. Top 10 Content Producing Countries



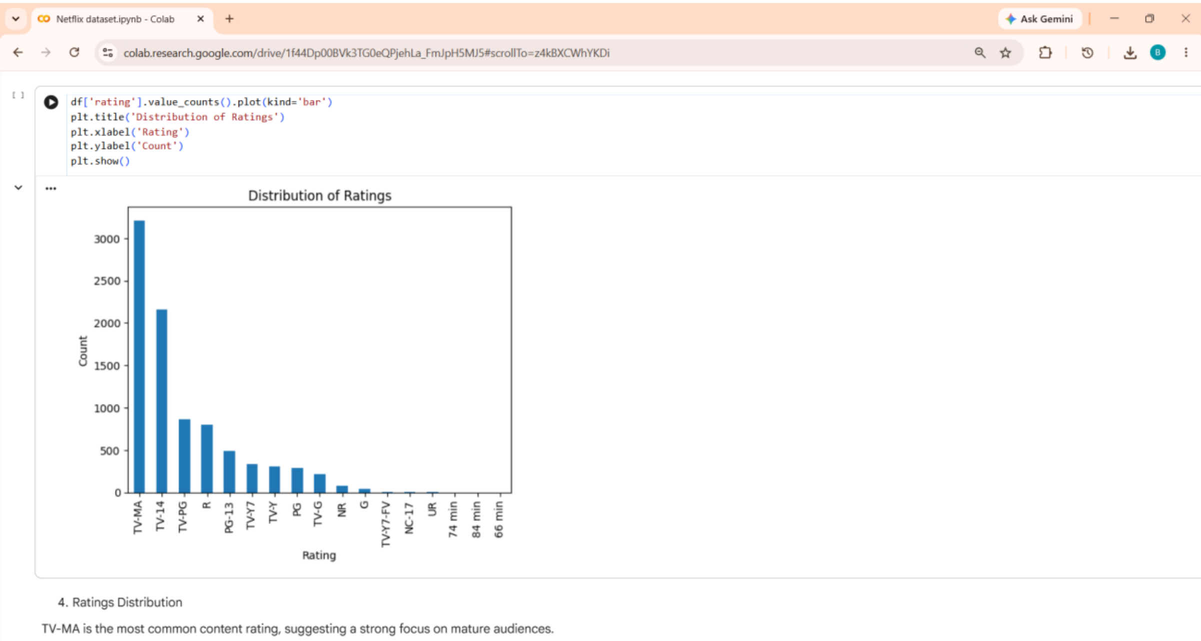
Observation: The United States contributes the highest number of titles on Netflix, followed by several other major content-producing countries.

3. Netflix Content Released by Year



Observation: Netflix content releases increased rapidly after 2015, indicating substantial growth in content production and acquisition.

4. Distribution of Content Ratings



Observation: TV-MA is the most common content rating, suggesting a strong focus on mature audiences.

5. Top Genres on Netflix

Observation: International Movies and Drama-related genres are among the most popular categories available on Netflix.

6.Key Insights

1. Movies dominate Netflix's content library.
2. The United States is the leading contributor of Netflix content.
3. Content production and acquisitions increased significantly after 2015.
4. Mature-audience content forms a major portion of Netflix's catalog.
5. International and drama-related genres are highly popular among available titles.

7.Conclusion

The Exploratory Data Analysis of the Netflix dataset revealed several important trends and patterns. Movies form the majority of Netflix content, while content availability has grown significantly in recent years. The United States contributes the highest number of titles, and mature-audience content is particularly prevalent. Furthermore, international and drama-related genres play a significant role in Netflix's diverse content library. Overall, the analysis provides valuable insights into Netflix's content distribution and growth trends.